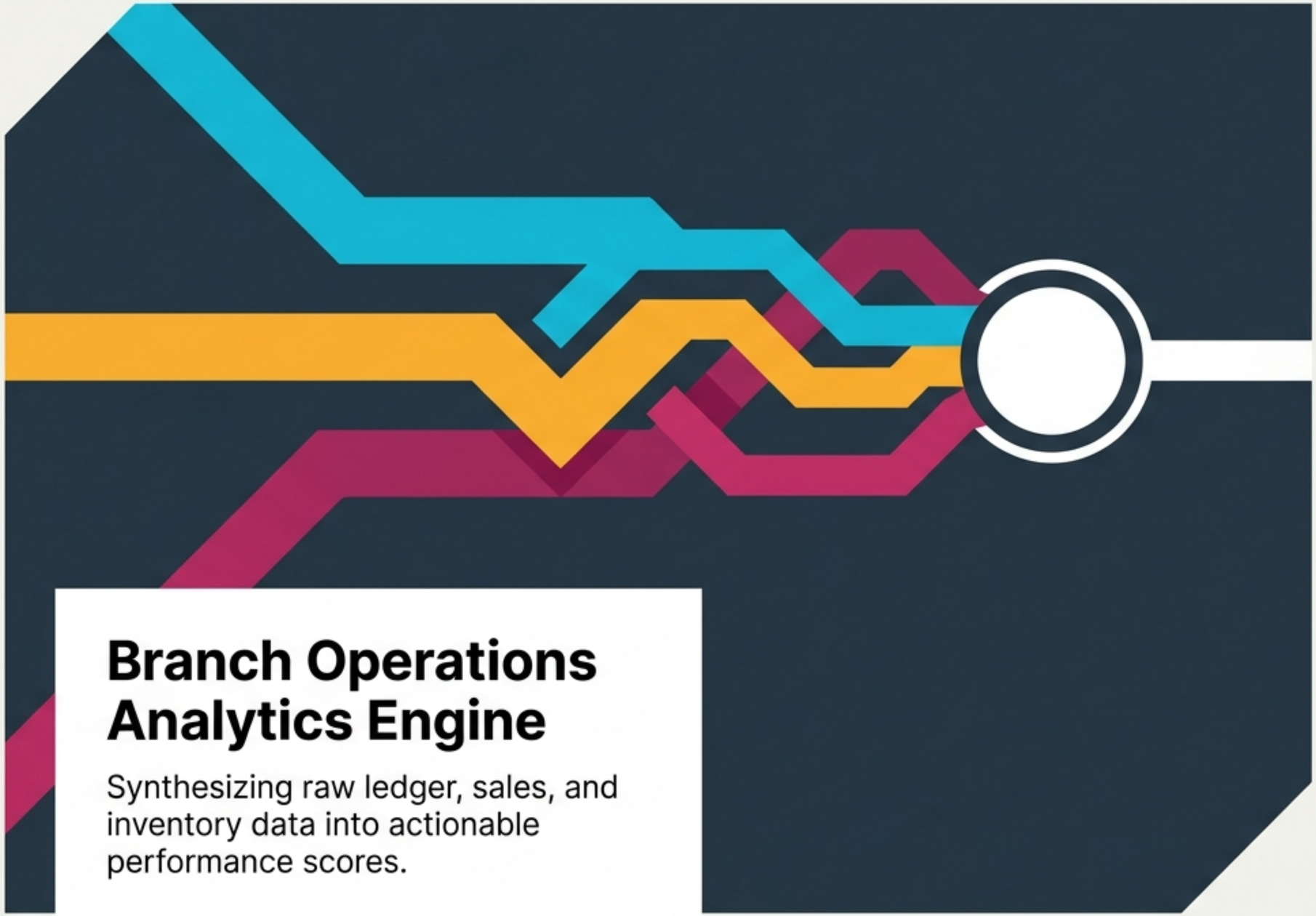
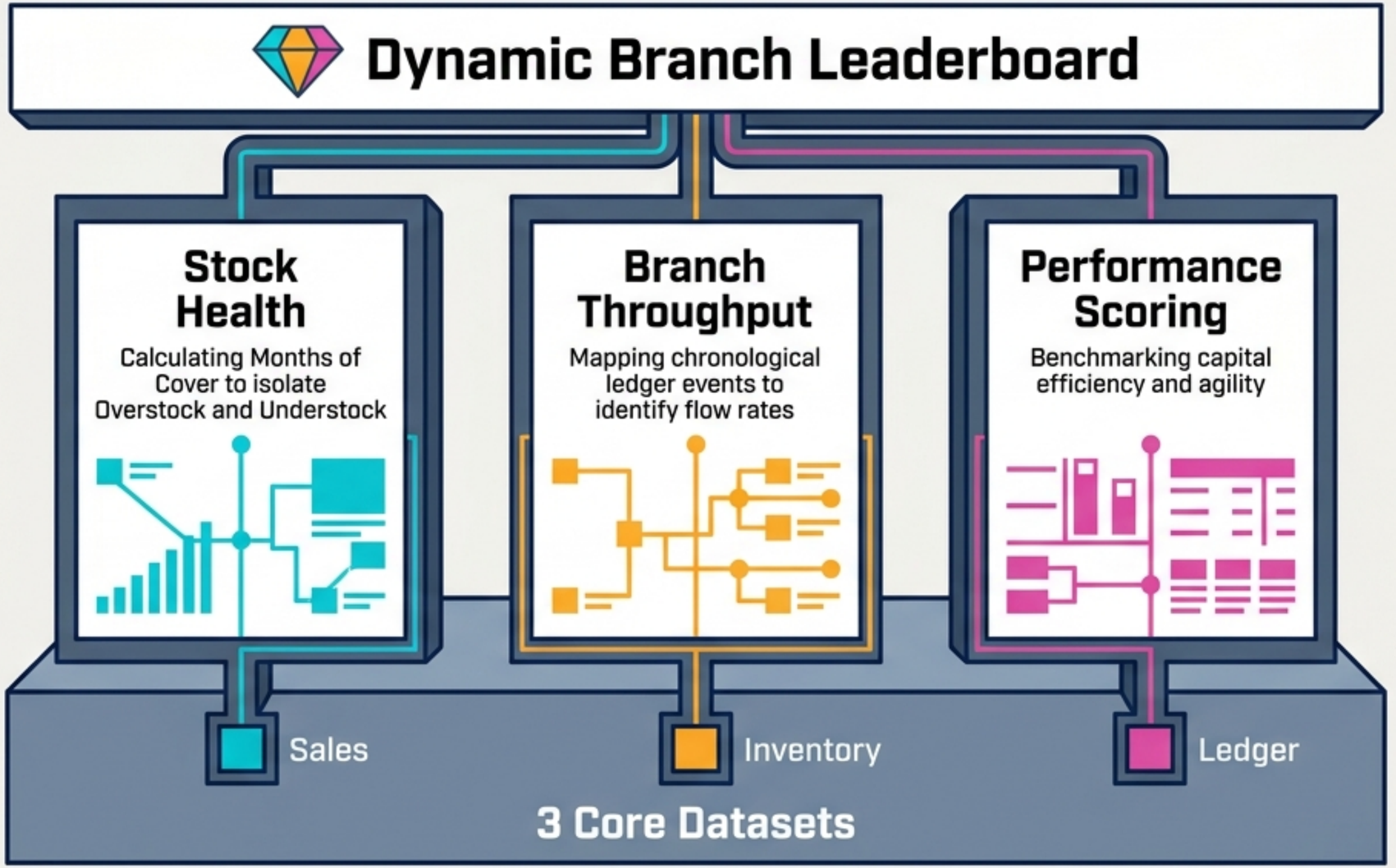


Branch Operations Analytics Engine

Synthesizing raw ledger, sales, and inventory data into actionable performance scores.

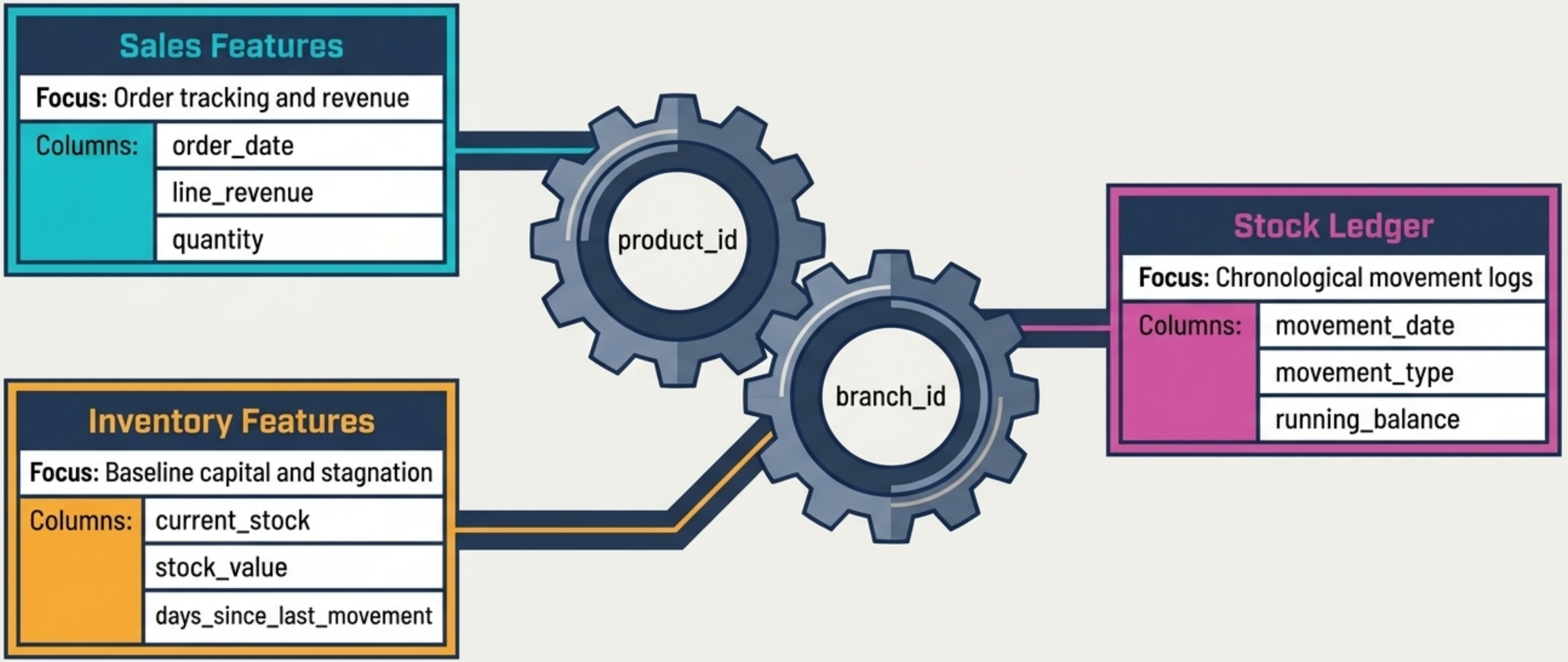
Gemini Notebook

Constructing a unified performance standard from three diagnostic pillars



Gemini Notebook

Interlocking three unique timelines to build a complete operational picture



Gemini Notebook

Pillar 1: Diagnosing stock health by aligning current inventory with historic demand

Avg Monthly Demand

Total Sales Quantity ÷ Months = Avg Monthly Demand

Months of Cover

Current Stock ÷ Avg Monthly Demand = Months of Cover

Stock Health Spectrum



Gemini Notebook

Over 92% of evaluated capital sits within the healthy 1-6 month cover window, providing a stable foundation to evaluate specific branch execution.

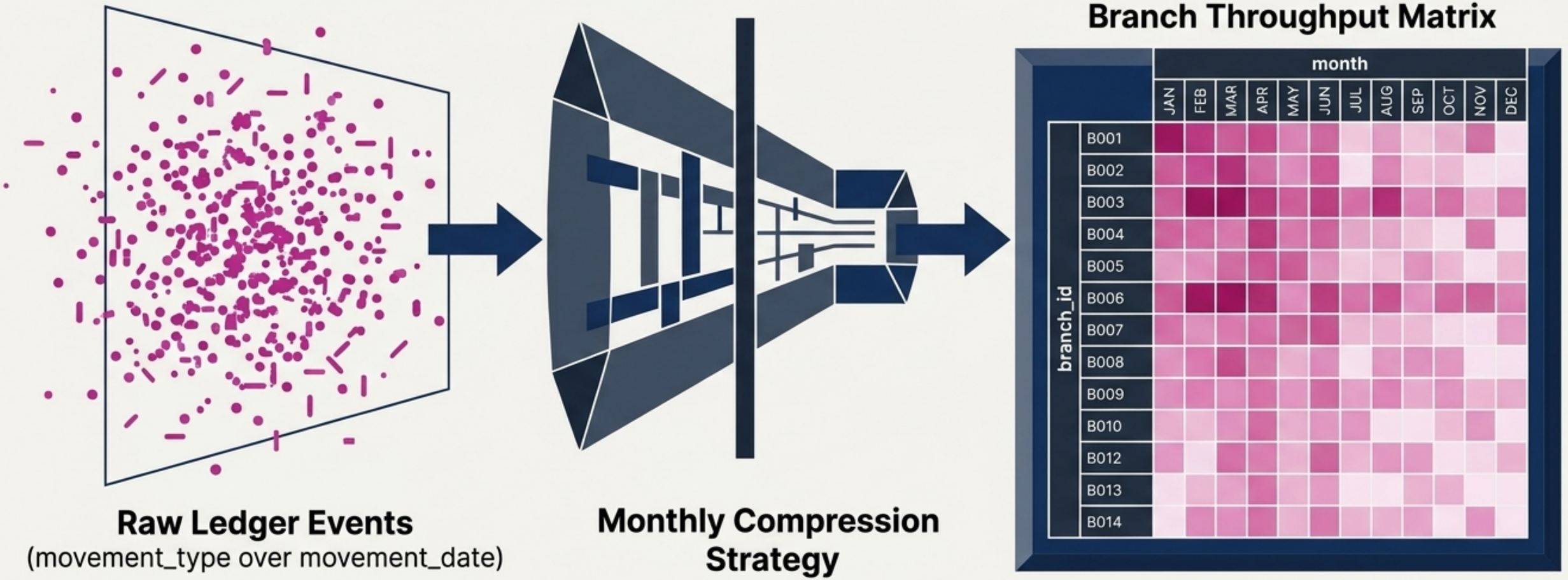
Healthy (1-6 months)
$443,404,250
155 item categories

Understock (<1 month)
$34,967,320
25 item categories

Gemini Notebook

Pillar 2: Structuring chronological ledger events to map branch flow rates

This transformation matrix isolates bottlenecks and identifies which branches move stock most efficiently month-over-month.



Raw Ledger Events
(movement_type over movement_date)

Monthly Compression Strategy

Gemini Notebook

Pillar 3: Isolating stagnant capital to penalize operational friction

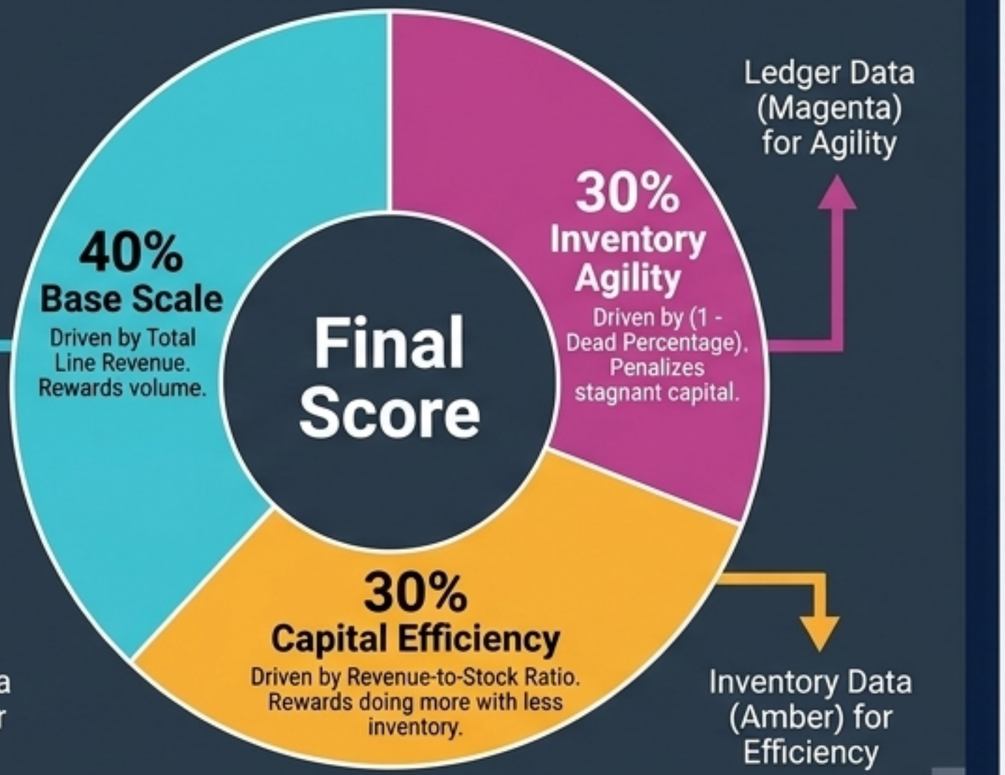


The top 6 performing branches successfully maintained $0 in dead value, completely avoiding agility penalties.

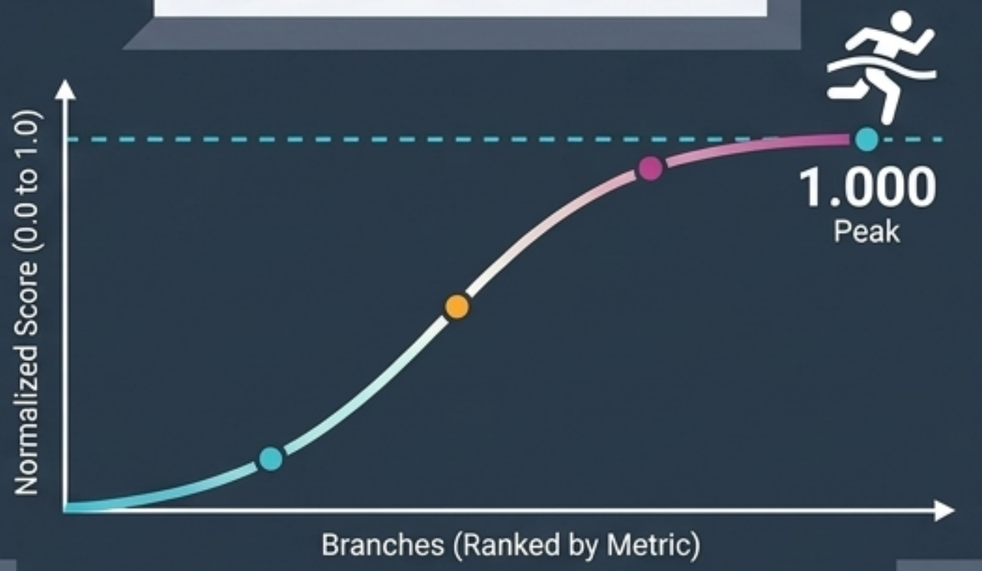
Gemini Notebook

The Performance Formula balances total scale with capital efficiency

The Weighted Engine



The Normalization Curve

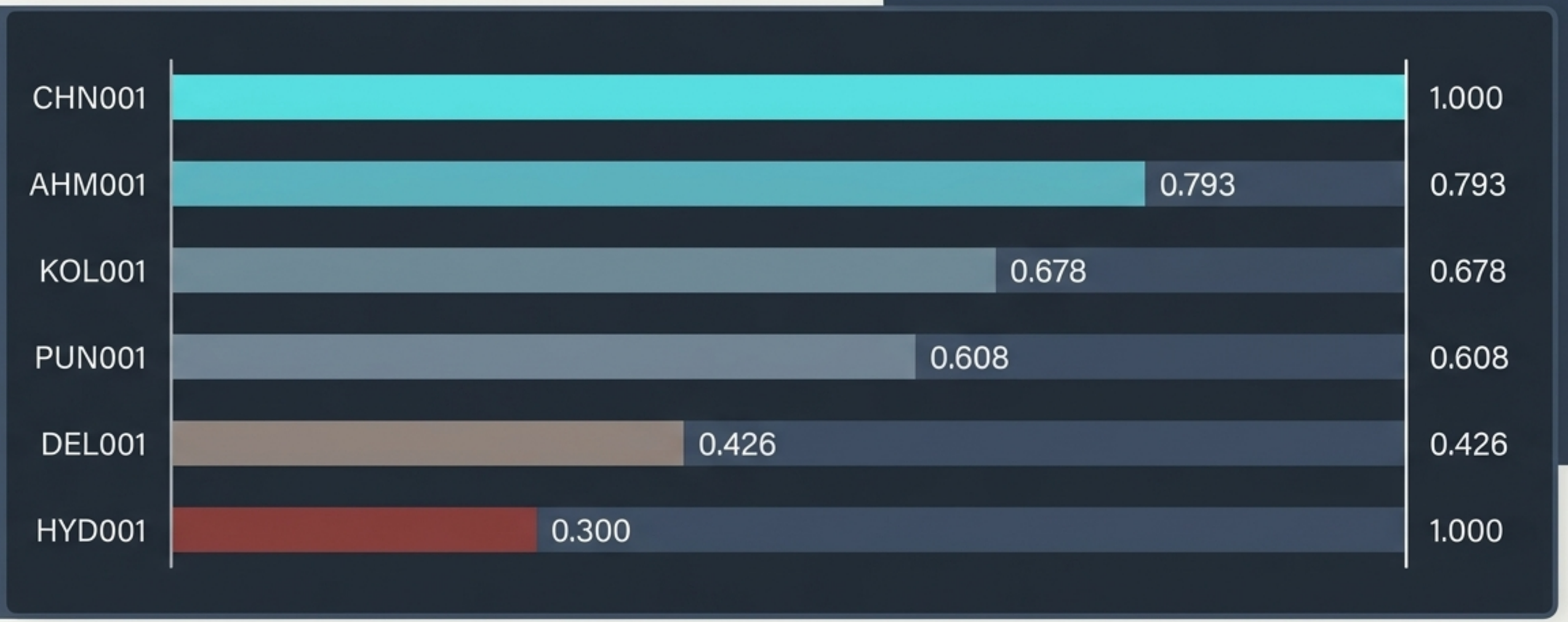




Min-Max Normalization: Branches are graded dynamically. The top performer in a metric sets the standard (1.0), and all others are measured relative to that peak.

Gemini Notebook

The Final Output:
Dynamic Branch Leaderboard



Gemini Notebook

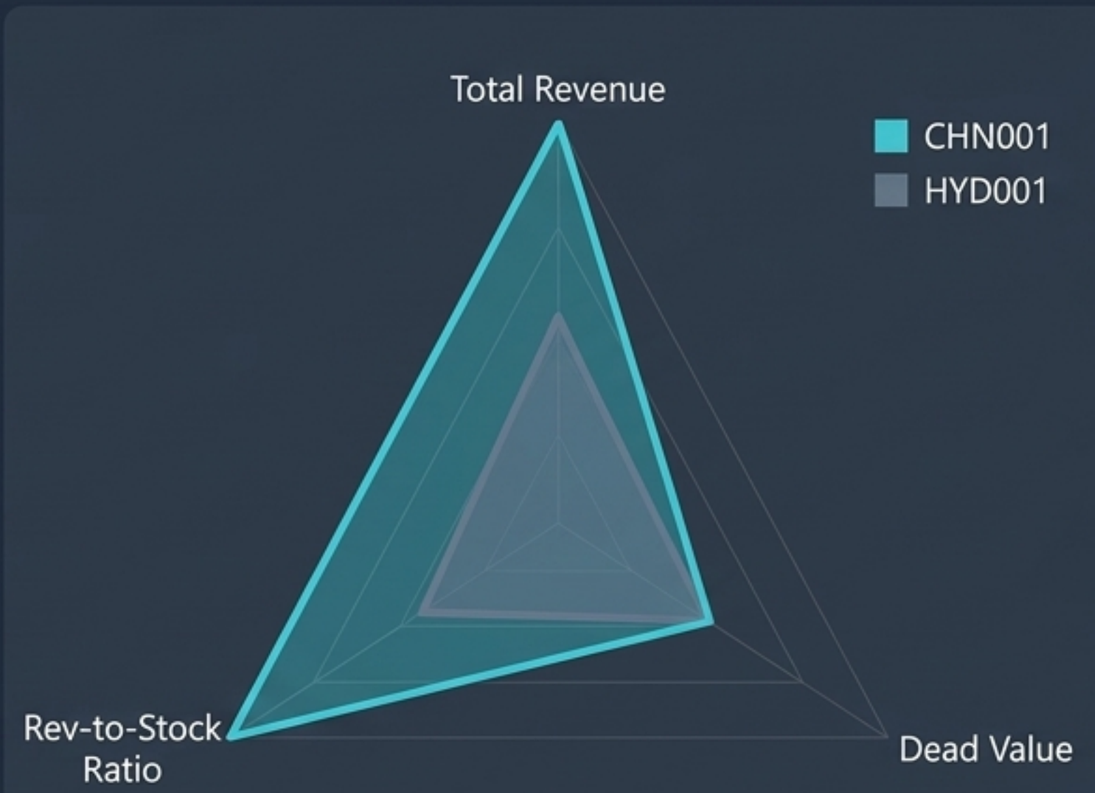
Diagnostic Contrast: Why CHN001 achieves the perfect 1.0 standard

CHN001 (#1)

Total Revenue:
$5.19B
Highest volume

Rev-to-Stock Ratio:
62.458
Highest efficiency

Dead Value:
0.0%



HYD001 (#6)

Total Revenue:
$3.45B
Lowest volume

Rev-to-Stock Ratio:
44.483
Lowest efficiency

Dead Value:
0.0%

CHN001 does not just sell the most; it generates its $5.19B revenue with highly efficient stock utilization, maximizing every dollar of inventory on hand.

Gemini Notebook

Deploying the Analytics Engine across operational teams



stock_status.csv



branch_throughput.csv



branch_scores.csv

Procurement

Action: Rebalance the $34.9M in understock risk

Logistics

Action: Optimize regional supply chain flow rates

Regional Management

Action: Establish KPI targets based on CHN001's 1.000 benchmark

Gemini Notebook